

Safety, Health, and Environmental risk assessment

Guidance: Complete sections 1.1-4.1 in order before writing summary

1.1 Activity/task

1.2 Relevant documents

1.3 Project/Lab location

Aerosol Extraction System - 4th Year MEng Group Project

MechSpace

Review date

MechSpace

2.1 Hazard category What might cause harm?	3.1 Hazard category	3.2 Groups affected			3.3 Nature of risk What might go wrong?	3.4 Risk before controls T / I	4.1 Preventative/control measures How do you stop it going wrong?	4.2 Risk after controls T / I
		P	O	M				
1 Fall from height		Y			Experimental rig will use aluminium profiles, in addition to an airbrush, mist generator and a conveyor belt. Together, these items can prove bulky and heavy to carry around.	T	Divide components into groups, then place in bags/containers for transport. Team members to aid as necessary when moving parts around.	T
2 Fall on same level								
3 Fall Down stairs/steps		Y			Use of saws and laser cutter for aluminium profile and acrylic panes. Injury possible if used improperly.	T	Appropriate PPE and precaution when cutting profiles. MechSpace mentors notified when a laser cutting job is conducted.	T
4 Struck by moving object			Y		Airbrush and mist generators used to simulate ink/water droplets and misting. Water used is dyed.	T	Stand far enough away to minimise risk of inhaling mist. Masks worn if necessary. No spraying paint is used.	T
5 Struck by falling object		Y			Conveyor belt used to simulate printer substrate requires a power source. Electric motor and some wiring may be exposed to moisture from misting.	T	Internal electronics will not be modified and will be used conventionally. Motor and wiring will be wrapped in plastic. If insufficient, de-ionised water will be used	T
6 Struck/trapped by something collapsed/overturned					Air suction and/or induced air flow will be introduced to rig. Vacuum is used to provide the suction. Pressure of suction/induced airflow is low as it is a domestic vacuum.	T	The vacuum will only be turned on when fully secured and ready.	T
7 Trapped between objects		Y	Y		Conveyor belt generates a fair amount of noise when in use. Extraction system prototype uses a vacuum which is also expected to produce noise from suction/air flow.	T	Use earbuds when prolonged use is expected. Other people in area also notified of use and advised to wear earbuds.	T
8 Step on/struck against stationery object					Experimental rig is easily put together/broken down. Aluminium profiles could cut on its edges.	T	Appropriate PPE worn when handling.	T
9 Manual handling		Y			The generation of mist (small water droplets) may cause condensation and high humidity over extended operation.	I	Conduct testing in well-ventilated area. Masks worn if necessary.	T
10 Contact with tool/equipment/machinery			Y		Vacuum will require mains power.	T	Vacuum cabling will be kept neat to avoid tripping hazards. The vacuum electronics will not be modified and will be used conventionally.	T
11 Contact with flying particle		Y			Conveyor belt has rotating sprockets on the side.	T	Conveyor belt will be turned off when not in use. It is inside an enclosed box during use.	T
12 Contact with low voltage electricity (<60V)								
13 Contact with high voltage electricity (60-800V)								
14 Contact with/exposure to air/water pressure								
15 Contact with/exposure to heat/cold								
16 Contact with/exposure to hazardous substance								
17 Exposure to noise								
18 Exposure to vibration								
19 Exposure to rotating parts								
20 Repetitive motion/action								
21 Collision								
22 Fire/flammable Atmosphere								
23 Explosion								
24 Assembling and Disassembling								
25 Asphyxiation								
26 Adverse weather conditions								
27 Sharp objects								
28 Oxygen enrichment								
29 Exposure to fumes or dust								
30 Exposure to Carbon or glass fibres								
31 Exposure to Arc flash								
32 Environmental threat								
33 Other State								

Quantifying the risk		No injury, damage or env impact	Minor injury, damage or env impact	Major injury, damage or env impact	Fatality, building loss, catastrophic env impact
		0	1	2	3
Almost no probability	0	L	L	L	L
A small probability	1	L	L	M	H
A high probability	2	L	M	H	H
Almost certain	3	L	H	H	H

Person completing the assessment
Academic

T Tolerable

I Intolerable

5.1 SIGN OFF

Name	Signature	Date
Name(s)	Ralf den Hartog	21/1/2026
Staff Name		

Key: S Safety; H Health; E Environment; L Low risk; M Medium risk; H High risk; P People carrying out the activity/task; O Others working in the vicinity; M Members of the public

Notes:
1. Where young persons or expectant mothers are involved in the activity, ensure that any additional controls are put in place in accordance with local procedures
2. Where a hazard is identified that is not listed, enter 33 in the Hazard Category column and describe under 'Nature of risk'